

WEEK 3: ADVANCED SECURITY AND FINAL REPORTING

1. Basic Penetration Testing

2. Set Up Basic Logging

```
const winston = require('winston');

const logger = winston.createLogger({
  transports: [
    new winston.transports.Console(),
    new winston.transports.File({ filename: 'security.log' })
  ]
});

pool.query('SELECT * FROM admin WHERE username = ?', [username], async function(err, rows)
{
  if (err || rows.length === 0) {
    return res.status(400).send("User not found");
  }

  logger.info(`Login attempt by ${username} at ${new Date().toISOString()}`);
```

3. Create a Simple Checklist

```
const https = require('https');

const fs = require('fs');

const httpsOptions = {
  key: fs.readFileSync('key.pem'),
  cert: fs.readFileSync('cert.pem')
};

const httpsServer = https.createServer(httpsOptions, app);

httpsServer.listen(8443, () => {
  console.log("HTTPS Server Listening on port 8443");
});

console.log("Server Listening on port 8443");
```